

Roll No.....

ODD Mid Semester Examination 2025

Name of the Program: B. Tech(ECE) (ESR and IoT)

Semester: IIIrd

Name of the Course: Fundamentals of Computer Organization Applications

Course Code: **TEC-359**

Time: 90 Minutes

Maximum Marks: 50

Note:

- (i) Answer **all the questions** by choosing **any one of the sub questions**.
- (ii) Each question carries 10 marks

Q1	(10 marks)	CO1
(a)	Explain computer structure briefly using suitable diagram and the functions of computer structure.	
OR		
(b)	Differentiate between Computer Organization and Computer Architecture.	
Q2	(10 marks)	CO1
(a)	Draw and explain the Functional View of Computer.	
OR		
(b)	Differentiate the processors architectures according to their memory usage technique.	
Q3	(10 marks)	CO2
(a)	Describe the Instruction and the internal register organization of Computer.	
OR		
(b)	Explain RISC and CISC along with all the advantages and disadvantages.	
Q4	(10 marks)	CO2
(a)	Write a program that can evaluate following expression: $X = (A * B) \times (C * C)$ Using zero, one, two, three address instructions. Assume that the processor has Load, Store, Multiply, and Add instructions, and that all values fit in the registers in processor.	
OR		
(b)	Explain concept of addressing modes used and explain each by example.	
Q5	(10 marks)	CO3
(a)	Explain the addition/subtraction algorithm with example.	
OR		
(b)	Explain Booth's multiplication algorithm and hence multiply $(6) \times (-7)$ using $n=4$	CO3

